

# ZHENZHI TAN

✉ BoeingART · in Zhenzhi Tan

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## 🎓 EDUCATION

CBMS, Department of Chemistry, **Tsinghua University**

Sept. 2024 – Present

*Ph.D. Student in Organic Chemistry (AI4Science)* · Advisor: Prof. Sanzhong Luo

Department of Chemistry, **Tsinghua University**

Sept. 2020 – July 2024

*B.S. in Chemistry* · **GPA: 3.92/4.00 (Rank: 2/25)** · Outstanding Graduate of Tsinghua University

## 🔬 RESEARCH EXPERIENCE

**Data-Driven Discovery & Active Learning for Cooperative Catalysis**

Mar. 2024 – Present

*Co-first Author / Project Leader (NSFC Ph.D. Project)* *Nature Synth.* (2025) [📄 Paper] [🔗 Code]

- **Active Learning:** Developed Bayesian Optimization (BO) algorithms tailored for small-sample, high-noise organic synthesis, addressing search challenges in high-dimensional chemical spaces.
- **Closed-loop Workflow:** Implemented an automated DFT descriptor extraction pipeline integrated with BO, enabling a “Design-Feature Extraction-Optimization” one-stop platform.
- **Efficiency:** Achieved global optimization of a 1.8-million reaction space within only 63 experiments, significantly accelerating the discovery of novel synergistic catalytic reactions.

**Ullmann Reaction Prediction & SynAD Reliability Framework**

Sept. 2023 – Oct. 2025

*First Author / Project Leader (NSFC Undergraduate Project)* *Angew. Chem. Int. Ed.* (2026) [📄 Paper] [🔗 Code]

- **Feature Engineering:** Built a high-accuracy Ullmann reaction yield prediction model by integrating physical organic chemistry knowledge with automated DFT-based descriptor computation.
- **Uncertainty Assessment:** Proposed the **Synthetic Applicability Domain (SynAD)** framework to address the “extrapolation trap” of AI in synthesis, establishing a confidence mechanism for unknown chemical spaces.
- **Validation:** Benchmarked the framework on 7 datasets, proving that SynAD effectively filters unreliable predictions and enhances the robustness of AI-driven synthesis planning.

**Computer Vision for Polymer Miscibility Prediction**

May 2022 – July 2023

*Second Author* *J. Chem. Inf. Model.* (2023) [📄 Paper] [🔗 Code]

- Constructed a Deep Convolutional Neural Network (CNN) to automatically extract high-level features from SEM microscopic images of polymer blends.
- Enabled rapid characterization of miscibility for polymers, replacing traditional labor-intensive manual analysis.

## ⚙️ SKILLS

- **Programming:** Python (PyTorch, Scikit-learn, Numpy, Pandas), C++, SQL, Linux/Bash, Git.
- **AI4Chem:** Cheminformatics (RDKit, DeepChem), DFT calculation (Gaussian, ORCA, and xTB).

## 🏆 HONORS & AWARDS

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|--------------------------------------------------------------------------------------|------|
| • Participant, 74th Lindau Nobel Laureate Meeting                                    | 2025 |
| • Best Poster Award, <i>Nature Conference on Automation for Chemistry</i>            | 2025 |
| • Best Poster Award, <i>Intl. Conf. of Computational Organic-synthesis Catalysis</i> | 2025 |
| • Outstanding Graduate of Tsinghua University (Top honor for graduates)              | 2024 |
| • T. P. Hou Award                                                                    | 2024 |
| • National Scholarship                                                               | 2022 |
| • Gold Medal, 33rd Chinese Chemistry Olympiad (CChO)                                 | 2019 |

## 📄 SELECTED PUBLICATIONS

- [1] **Z. Tan**, Q. Yang, L. Zhang, & S. Luo. *Angew. Chem. Int. Ed.* **2026**, e23874.
- [2] L. Cheng<sup>‡</sup>, **Z. Tan<sup>‡</sup>**, Z. Jia<sup>‡</sup>, et al. *Nature Synth.* **2025**, 10.1038/s44160-025-00944-y.
- [3] **Z. Tan**, Q. Yang, & S. Luo. *Org. Chem. Front.* **2025**, 12, 2759-2776.
- [4] Z. Liang, **Z. Tan**, et al. *J. Chem. Inf. Model.* **2023**, 63, 5971-5980.